

Please note that this document only shows sample inputs and expected outputs to illustrate general behaviour. You are expected to design and execute additional test cases—including boundary, edge, and invalid conditions. Your test suite must include multiple input sets to thoroughly verify that all functionalities and path logic of your application operate correctly.

Volcano Monitoring

This is a sample input data and the corresponding output.

Volcano Monitoring & Eruption Risk Evaluation Summary

Volcano Name	Type	Core Attributes	Eruption Risk Score	GeoNet Alert Level	Public Warning Triggered?
Ruapehu	Stratovolcano	- Gas Pressure: 110.0 - Avg Seismic: 80.0	95.0	Level 5	Yes (Broadcast Warning)
Ngāuruhoe	Stratovolcano	- Gas Pressure: 30.0 - Avg Seismic: 20.0	25.0	Level 1	No
Taupō	Caldera	- Path: Root → V1 → S1 - Max Pressure: 45 (≤ 100)	90.0	Level 5	Yes (Broadcast Warning)
Rotorua	Caldera	- Path: Root → V2 → S4 - Blocked Node: V2 (120 > 100)	20.0	Level 1	No
Rangitoto	Shield	- Lava Flow Rate: 80.0 - Ground Inflation: 65.0	71.0	Level 3	Yes (Broadcast Warning)

Volcanic Hazard Ranking (Most to Least Dangerous)

- Ruapehu** — Risk Score: 95.0 (Level 5: Major Volcanic Eruption)
- Taupō** — Risk Score: 90.0 (Level 5: Major Volcanic Eruption)
- Rangitoto** — Risk Score: 71.0 (Level 3: Minor Volcanic Eruption)
- Ngāuruhoe** — Risk Score: 25.0 (Level 1: Minor Volcanic Unrest)
- Rotorua** — Risk Score: 20.0 (Level 1: Minor Volcanic Unrest)

GeoNet Active Warning Broadcast Log (Stack Behavior)

- Top of Stack (Most Recent Alert / First to Undo):** Rangitoto (Level 3)
- Second Most Recent Alert:** Taupō (Level 5)
- Bottom of Stack (Oldest Active Alert):** Ruapehu (Level 5)

Sample Software Testing Table

Feature / Module	Test Description	Preconditions	Input Data	Expected Result	Actual Result	Status
Risk Calculation	Verify Stratovolcano risk score calculation	User logged in, GeoNet system active	gasPressure = 110.0 avgSeismic = 80.0	Risk score = 95.0, GeoNet Level 5	Risk score = 95.0, GeoNet Level 5	PASS
Vent Traversal	Validate Caldera backtracking on blocked vent	Vent 2 pressure set to 120	Trigger magma path exploration from Root	System backtracks from Vent 2, returns Risk = 20.0	System backtracks from Vent 2, returns Risk = 20.0	PASS
Emergency Broadcast	Trigger warning alert when risk score ≥ 75	System alert stack initialized	riskScore = 95.0 (Ruapehu)	Public warning broadcasted; appended to active alert stack	Public warning broadcasted; appended to active alert stack	PASS
Undo Warning	Verify LIFO popping of false alarm warning	At least 1 active warning in stack	Execute undoLastAlert()	Most recent warning (Ruapehu) removed from active stack	Warning removed from active stack	PASS

Dragon’s Hoard Manager

This is a sample input data and the corresponding output.

Lair Hoard & Security Evaluation Summary

Entity Type	Item / Squad Name	Core Attributes	Calculated Value	Security Threat Level	Required Security Action
Gemstone	Imperial Ruby	- Weight: 15.0 carats - Clarity: 9 - Avg Appraisal: 250	7,000	Level 4	Arm lethal traps
Gemstone	Flawless Emerald	- Weight: 2.0 carats - Clarity: 5 - Avg Appraisal: 40	540	Level 2	Watch status
Relic	Reachable Relic	- Path: $A \rightarrow B \rightarrow G \rightarrow R1$ - Max Danger: 80 (≤ 100)	250	Level 1	Log only
Relic	Unreachable Relic 1	- Path: $A \rightarrow D \rightarrow R2$ - Blocked Node: D ($110 > 100$)	10,000	Level 5	Full lair lockdown
Relic	Unreachable Relic 2	- Path: $A \rightarrow C \rightarrow E \rightarrow H \rightarrow R3$ - Blocked Node: E ($160 > 100$)	10,000	Level 5	Full lair lockdown
Mercenary	Iron Golem Legion	- Guard Count: 10 - Daily Rate: 25	7,500	Level 0	No action
Mercenary	Wyrmguard Vanguard	- Guard Count: 4 - Daily Rate: 15	1,800	Level 0	No action

Sorted Hoard Inventory (Highest to Lowest Value to the Dragon)

- Unreachable Relic 1 — 10,000 (Threat Level 5: Full Lair Lockout)
- Unreachable Relic 2 — 10,000 (Threat Level 5: Full Lair Lockout)
- Imperial Ruby — 7,000 (Threat Level 4: Arm Lethal Traps)
- Flawless Emerald — 540 (Threat Level 2: Watch Status)
- Reachable Relic — 250 (Threat Level 1: Log Only)

Mercenary Deployment Order (Stack Behavior)

- Top of Stack (Most Recently Deployed / First to Dismiss): Wyrmguard Vanguard
- Bottom of Stack (First Deployed / Last to Dismiss): Iron Golem Legion

Sample Software Testing Table

Feature / Module	Test Description	Preconditions	Input Data	Expected Result	Actual Result	Status
Gemstone Valuation	Calculate value and security threat level for a high-value gemstone	System initialized with standard appraisal formulas	weight = 15.0 clarity = 9 avgAppraisal = 250	Value = 7000, Threat Level 4 ("Arm lethal traps")	Value = 7000, Threat Level 4 ("Arm lethal traps")	PASS
Relic Reachability	Recursively verify reachable relic path	Tunnel network loaded with open path ($A \rightarrow B \rightarrow G \rightarrow R1$)	Target = Reachable Relic (Chamber G, max danger 80)	Traversal succeeds, Relic Value = 250, Threat Level 1 ("Log only")	Traversal succeeds, Relic Value = 250, Threat Level 1 ("Log only")	PASS
Mercenary Valuation	Calculate mercenary squad value and check threat level exemption	Mercenary registration module active	guardCount = 10 dailyRate = 25	Value = 7500, Threat Level 0 ("No action / Replaceable")	Value = 7500, Threat Level 0 ("No action / Replaceable")	PASS
Squad Dismissal	Validate LIFO stack order when dismissing excess mercenaries	Two squads deployed: Iron Golem first, then Wyrmguard	Trigger dismissSquad()	Wyrmguard Vanguard dismissed first (Top of stack)	Wyrmguard Vanguard dismissed first (Top of stack)	PASS